Header Parameter Models

If you have a group of related **header parameters**, you can create a **Pydantic model** to declare them.

This would allow you to **re-use the model** in **multiple places** and also to declare validations and metadata for all the parameters at once. 😎

> **Note**
>
> This is supported since FastAPI version `0.115.0`. 🤓

Header Parameters with a Pydantic Model

Declare the **header parameters** that you need in a **Pydantic model**, and then declare the parameter as `Header`:

Python 3.10+

```python
from typing import Annotated

from fastapi import FastAPI, Header
from pydantic import BaseModel

app = FastAPI()


class CommonHeaders(BaseModel):
    host: str
    save_data: bool
    if_modified_since: str | None = None
    traceparent: str | None = None
    x_tag: list[str] = []


@app.get("/items/")
async def read_items(headers: Annotated[CommonHeaders, Header()]):
    return headers
```

Ask AI

🤓 **Other versions and variants**

Python 3.10+ - non-Annotated

Tip

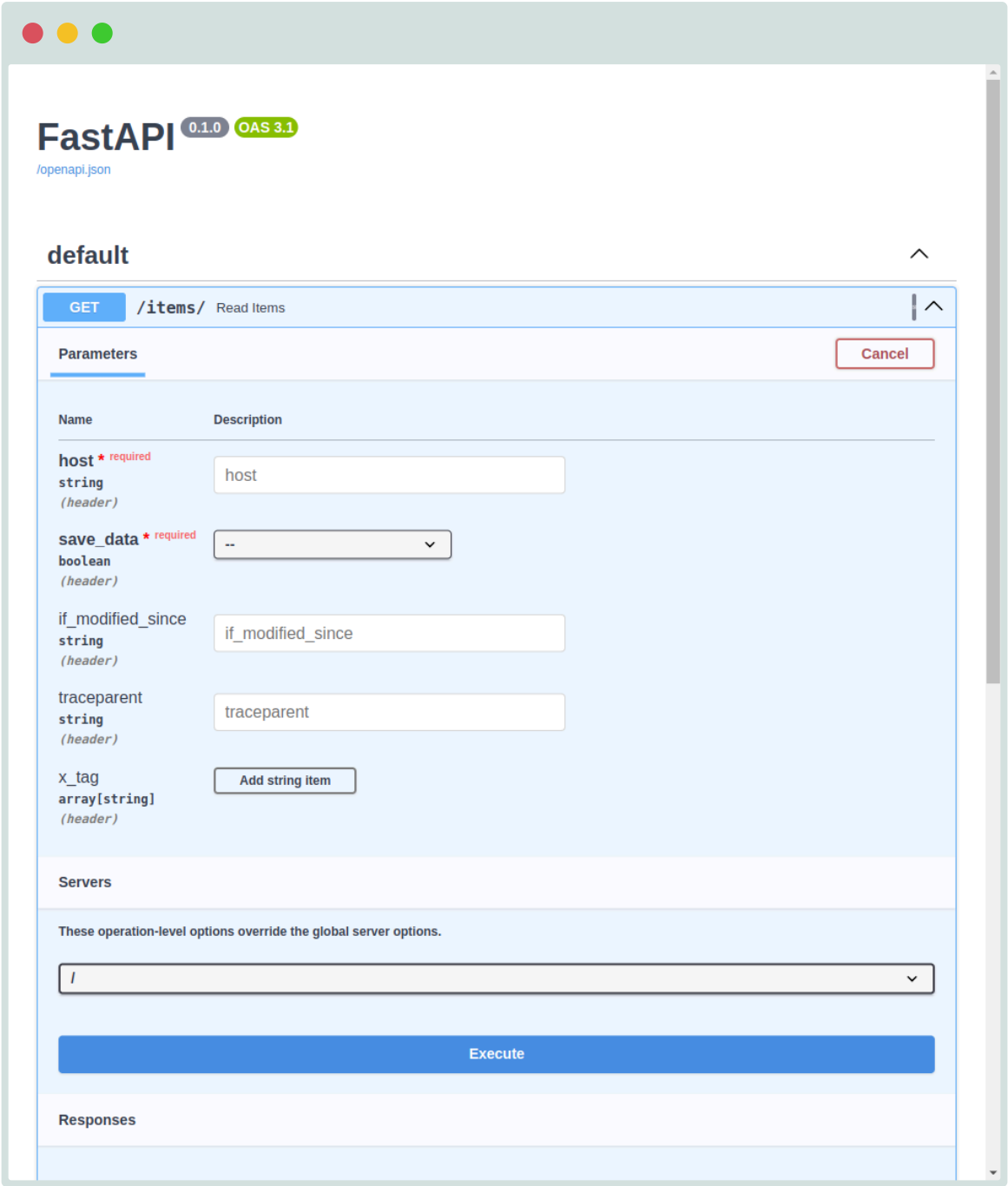
Prefer to use the `Annotated` version if possible.

```python
from fastapi import FastAPI, Header
from pydantic import BaseModel

app = FastAPI()


class CommonHeaders(BaseModel):
    host: str
    save_data: bool
    if_modified_since: str | None = None
    traceparent: str | None = None
    x_tag: list[str] = []


@app.get("/items/")
async def read_items(headers: CommonHeaders = Header()):
    return headers
```

FastAPI will **extract** the data for **each field** from the **headers** in the request and give you the Pydantic model you defined.

Check the Docs

You can see the required headers in the docs UI at `/docs`:



Forbid Extra Headers

In some special use cases (probably not very common), you might want to **restrict** the headers that you want to receive.

You can use Pydantic's model configuration to `forbid` any `extra` fields:

Python 3.10+

```
from typing import Annotated

from fastapi import FastAPI, Header
from pydantic import BaseModel

app = FastAPI()
```

```python
class CommonHeaders(BaseModel):
    model_config = {"extra": "forbid"}

    host: str
    save_data: bool
    if_modified_since: str | None = None
    traceparent: str | None = None
    x_tag: list[str] = []


@app.get("/items/")
async def read_items(headers: Annotated[CommonHeaders, Header()]):
    return headers
```

Other versions and variants

Python 3.10+ - non-Annotated

Tip

Prefer to use the `Annotated` version if possible.

```python
from fastapi import FastAPI, Header
from pydantic import BaseModel

app = FastAPI()


class CommonHeaders(BaseModel):
    model_config = {"extra": "forbid"}

    host: str
    save_data: bool
    if_modified_since: str | None = None
    traceparent: str | None = None
    x_tag: list[str] = []


@app.get("/items/")
async def read_items(headers: CommonHeaders = Header()):
    return headers
```

If a client tries to send some **extra headers**, they will receive an **error** response.

For example, if the client tries to send a `tool` header with a value of `plumbus`, they will receive an **error** response telling them that the header parameter `tool` is not allowed:

```
{
    "detail": [
        {
            "type": "extra_forbidden",
            "loc": ["header", "tool"],
            "msg": "Extra inputs are not permitted",
            "input": "plumbus",
        }
    ]
}
```

Disable Convert Underscores

The same way as with regular header parameters, when you have underscore characters in the parameter names, they are **automatically converted to hyphens**.

For example, if you have a header parameter `save_data` in the code, the expected HTTP header will be `save-data`, and it will show up like that in the docs.

If for some reason you need to disable this automatic conversion, you can do it as well for Pydantic models for header parameters.

Python 3.10+

```python
from typing import Annotated

from fastapi import FastAPI, Header
from pydantic import BaseModel

app = FastAPI()


class CommonHeaders(BaseModel):
    host: str
    save_data: bool
    if_modified_since: str | None = None
    traceparent: str | None = None
    x_tag: list[str] = []


@app.get("/items/")
async def read_items(
    headers: Annotated[CommonHeaders, Header(convert_underscores=False)],
):
    return headers
```

🤓 **Other versions and variants**

Python 3.10+ - non-Annotated

Tip

Prefer to use the `Annotated` version if possible.

```python
from fastapi import FastAPI, Header
from pydantic import BaseModel

app = FastAPI()


class CommonHeaders(BaseModel):
    host: str
    save_data: bool
    if_modified_since: str | None = None
    traceparent: str | None = None
    x_tag: list[str] = []


@app.get("/items/")
async def read_items(headers: CommonHeaders = Header(convert_underscores=False)):
    return headers
```

> **Warning**
>
> Before setting `convert_underscores` to `False`, bear in mind that some HTTP proxies and servers disallow the usage of headers with underscores.

Summary

You can use **Pydantic models** to declare **headers** in **FastAPI**. 😎